

The water cycle

Let's do an experiment:

Water cycle in a bag

Mark on the clear plastic bag each stage of the water cycle - use the diagram you have just completed to help you - ensuring you have the key stages of evaporation, condensation and precipitation marked on there. If you want to add houses, trees etc you can.

Next, fill a cup with regular tap water. Then add a couple of drops of blue food colouring to it to make it stand out. Swirl the cup of water a few times to mix the colouring until it's a nice, solid blue.

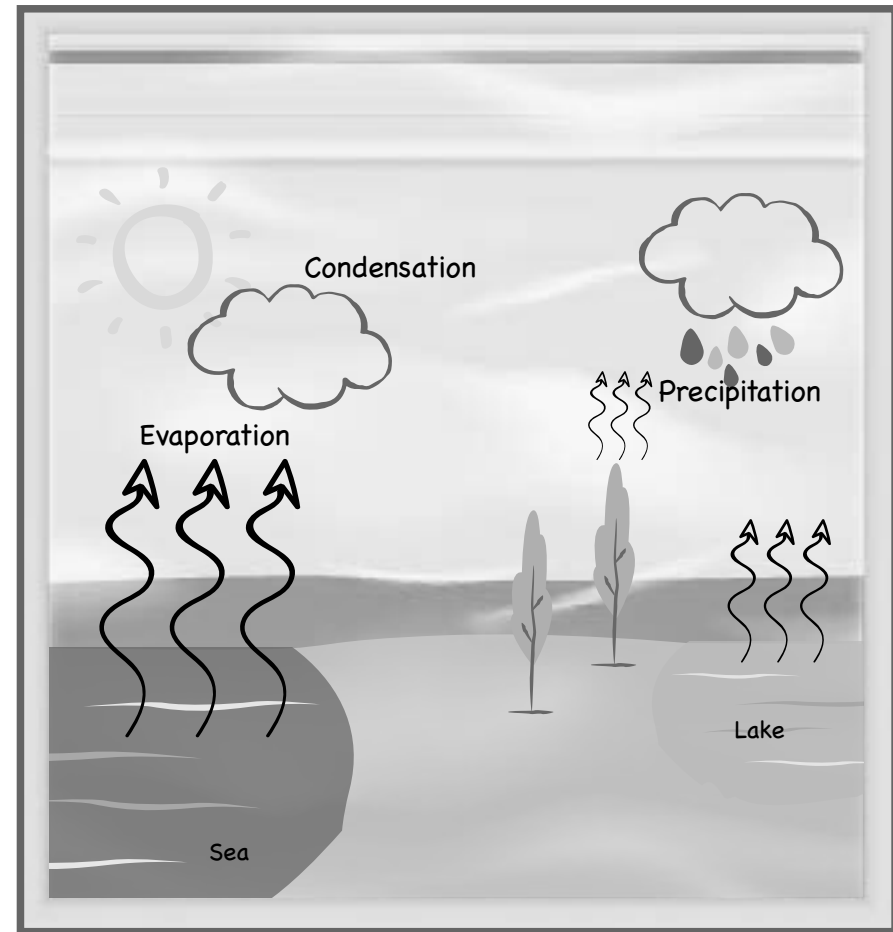
Now, carefully pour the water into the plastic bag and zip it closed. Make sure it's nice and tight! You don't want any water to be able to escape.

Once your water has been added to the bag and it's sealed up, it's time to hang it up on a window! Pick a window that gets a lot of sunshine for best results.

Cut two long pieces of tape and stick them to the corners of your bag. Then, press the tape tightly to the window - you don't want your bag to fall down!

Wait a couple of hours and check on your experiment. You should see a change in your bag between two hours and 1 day (depending on the amount of sun and the time of day you started).

Eventually, you will begin to see droplets of water sticking to the side of the bag. Some of these will be up high (in the clouds) while other droplets will be on their way back down (like rain).



Why is this happening?

It's because the water in the bag is being heated up against the sunny window. That water turns into a gas through a process called evaporation. In nature, evaporated water vapour goes into the atmosphere, but in our bag, it has nowhere to go, so it ends up sticking to the side of the bag, turning back into a liquid as condensation. That condensed water then slides back into the pool of water below as 'rain'.